

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2017-2018

INTRODUCTION TO SOFTWARE ENGINEERING

Time allowed: ONE HOUR

Candidates must NOT start writing their answers until told to do so

Answer ALL of the Questions

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

SECTION A**[25 Marks]****Software Development Scenario**

Your startup company have received a project from LRT Putrajaya about handling issues on ticket processing system. The scenario for this project is as follow: A customer service technician receives a telephone call, email, or other communication from a customer about a problem. Some applications provide built-in messaging system and automatic error reporting from exception handling blocks. The technician verifies that the problem is real, and not just perceived. The technician will also ensure that enough information about the problem is obtained from the customer. This information generally includes the environment of the customer, when and how the issue occurs, and all other relevant circumstances. The technician creates the issue in the system, entering all relevant data, as provided by the customer. As work is done on that issue, the system is updated with new data by the technician. Any attempt at fixing the problem should be noted in the issue system. Ticket status most likely will be changed from open to pending. After the issue has been fully addressed, it is marked as resolved in the issue tracking system. If the problem is not fully resolved, the ticket will be reopened once the technician receives new information from the customer.

1. Answer the following questions using the scenario given above.

- a) What is the difference between class diagram and state machine diagram? [2 Marks]
- b) Explain what is Use Case diagram? How use cases can help analysis team and development team in the project (2 reasons for each). [5 marks]
- c) Draw a use case diagram for the given above scenario. [7 Marks]
- d) List out 3 benefits of using sequence diagram in your design stage of your software development. [3 Marks]
- e) Draw a sequence diagram for the given above scenario. [8 Marks]

SECTION B**[25 Marks]**

2.

- a) State the advantages of using version control when working in a software development team that is distributed over multiple geographical locations. [4 Marks]

- b) list 5 functional requirements for the cash-dispensing function of a bank ATM machine. [5 marks]

- c)
 - i) Compare and contrast the Waterfall design methodology with eXtreme Programming (XP). [4 Marks]
 - ii) What are the benefits and drawbacks of both approaches to software design? [2 Marks]

- d)
 - i) Describe the Scrum development framework. [3 Marks]
 - ii) Why isn't Scrum used for all software development? [3 Marks]

- e) Explain the difference between "requirements definition" and "requirements specification"? [4 Marks]

THE END